

Project 2 Report – Systems Biology

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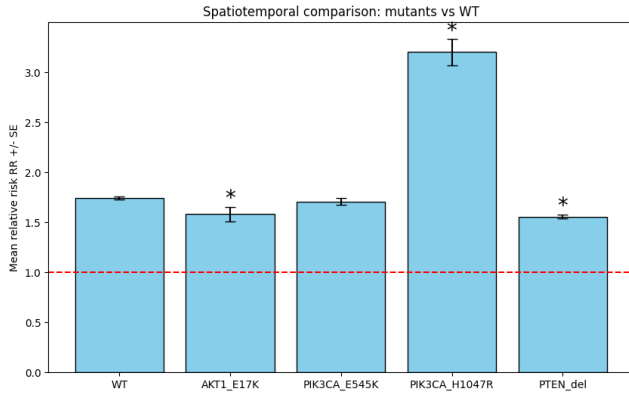
1 Part A

1.1 Task A1

1.1.1 Methods

Five cell lines were compared: WT, AKT1_E17K, PIK3CA_E545K, PIK3CA_H1047R, and PTEN_del. Mean relative risk (`mean_relative_risk`) was extracted from `group_level_summary.csv`; block-level values from `block_level_summary.csv`. Differences between WT and each mutant were tested with Mann–Whitney U tests with Bonferroni correction.

1.1.2 Key results



	mutation	mean_RR	std_error	p_value_vs_WT	significant
0	WT	1.742349	0.017193	NaN	False
1	AKT1_E17K	1.581421	0.071932	1.384587e-02	True
2	PIK3CA_E545K	1.707110	0.031885	7.341158e-01	False
3	PIK3CA_H1047R	3.200310	0.130643	1.225466e-08	True
4	PTEN_del	1.556633	0.021624	1.464234e-06	True

1.1.3 Interpretation

PIK3CA_H1047R showed a dramatically elevated relative risk ($RR \approx 3.20$), indicating exceptionally strong signalling coupling between neighbouring cells. Its kinase-domain location drives constitutive, strong PI3K activation, enabling rapid chain reactions across the cell population. The remaining mutants exceeded the reference line but remained near or below WT, suggesting that their chronic, weaker activation may induce habituation, reducing the efficiency of neighbour-driven signal transmission.

1.2 Task A2

1.2.1 Research question

Do PI3K–AKT pathway mutations alter the ERK relay timescale τ^* at which $RR(\tau)$ is maximised?

1.2.2 Methods

A jump event is a frame-to-frame increase $\Delta s \geq \theta$ (per-block 90th percentile). **Exposure:** ≥ 1 neighbour (radius 60, k-d tree) jumped at frame t . **Response at lag τ :** focal cell jumps at $t + \tau$.

$$RR(\tau) = \frac{P(\text{focal jumps at } t+\tau \mid \text{neighbour jumped at } t)}{P(\text{focal jumps at } t+\tau \mid \text{no neighbour jumped at } t)} \quad (1)$$

Lags $\tau \in \{0, \dots, 6\}$ frames (0–30 min) were evaluated; counts pooled across blocks. Three lines: **WT** ($n=24$), **AKT1_E17K** ($n=21$), **PTEN_del** ($n=27$).

1.2.3 Key results

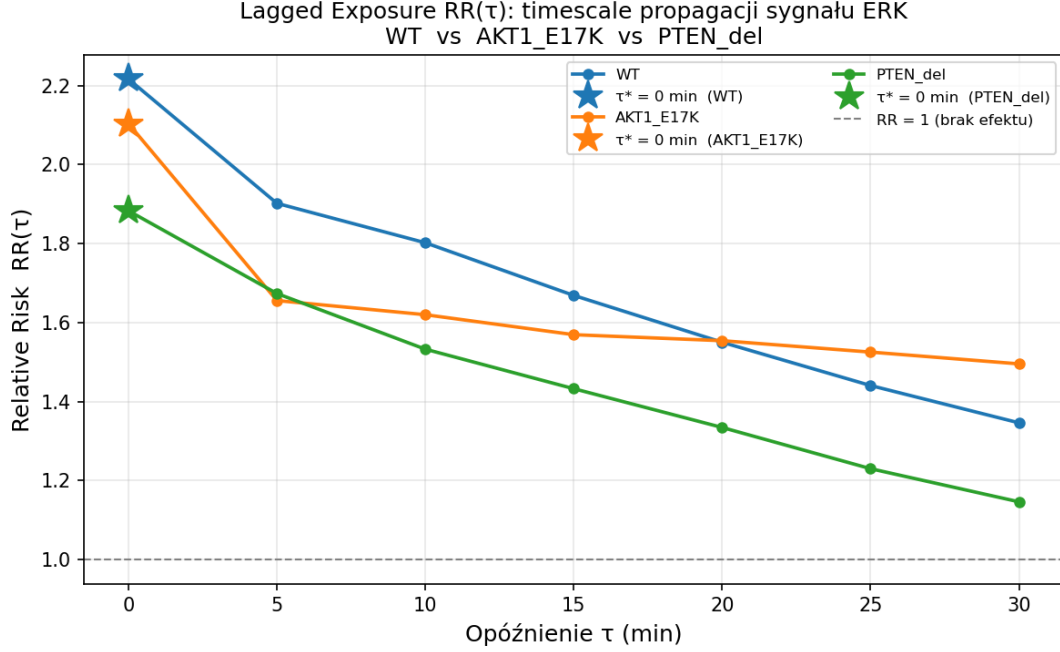


Figure 1: RR(τ) vs. lag (min) for WT (blue), AKT1_E17K (orange), PTEN_del (green). Stars: τ^* ; dashed line: RR = 1.

Table 1: Optimal lag τ^* and maximum RR per cell line.

Mutation	τ^* (fr)	τ^* (min)	max RR
WT	0	0	2.216
AKT1_E17K	0	0	2.103
PTEN_del	0	0	1.883

1.2.4 Interpretation

All three genotypes peak at $\tau^* = 0$, consistent with sub-5-min propagation or simultaneous response to a shared paracrine cue—indistinguishable at this frame rate. RR(τ) decays monotonically with lag in all lines. The gradient WT > AKT1_E17K > PTEN_del shows that PI3K–AKT hyperactivation *attenuates* spatial ERK coordination, likely by elevating autonomous jump rates and diluting the neighbour-driven signal. PTEN_del, acting furthest upstream, produces the strongest tonic noise and the lowest RR.

1.3 Task A3

1.3.1 Research question

Are the mutation-level RR differences robust to parameter choice?

1.3.2 Methods

One-at-a-time (OAT) sensitivity analysis; baseline $r=60$, $W=3$ fr, $q=0.90$. Ranges: $r \in \{30, 60, 90, 150\}$; $W \in \{1, 3, 6, 12\}$ fr; $q \in \{0.70, 0.80, 0.90, 0.95\}$. All five genotypes; Experiment 1 (19 blocks); counts pooled before computing RR.

1.3.3 Key results & interpretation

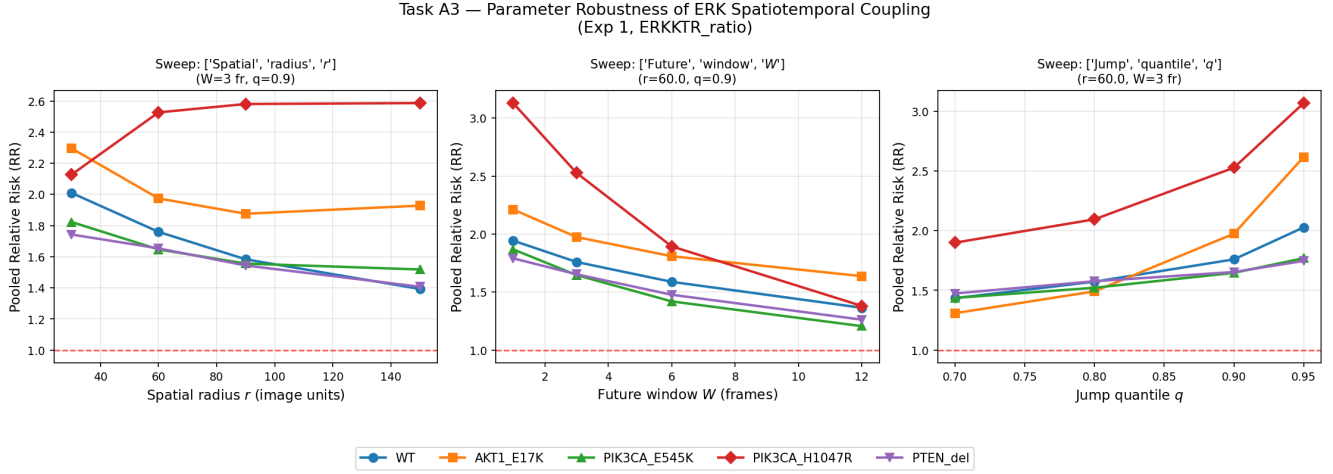


Figure 2: Pooled RR per mutation vs. swept parameter (Exp 1, ERKKTR_ratio). Dashed red: $RR = 1$. Left: radius sweep. Centre: window sweep. Right: quantile sweep.

PIK3CA_H1047R shows the highest RR under nearly all settings, but two rank inversions favour AKT1_E17K at extreme values. At $r=30$ it is overtaken by AKT1_E17K (2.30 vs. 2.13), possibly reflecting contact-dependent signalling dominant at very short range. For the future window, PIK3CA_H1047R peaks at $W=1$ fr ($RR=3.13$) but falls to 1.38 at $W=12$ fr, when AKT1_E17K (1.64) overtakes it—consistent with fast propagation diluted as autonomous jumps accumulate over 60 min. Across jump quantiles, PIK3CA_H1047R ranks first at all thresholds (1.90–3.07) with no inversion. The elevated coupling of PIK3CA_H1047R is therefore largely robust; the two exceptions arise only at extreme parameter values and are biologically interpretable.

1.4 Task B

We chose option 2 - open research question.

1.4.1 Introduction

Research question: **Does the size of a group of cells affect how effectively they transmit a signal to each other?**

In small groups of cells, the signal can quickly decay because once the wave reaches the vessel edge, it has nowhere else to go. In contrast, in large cell clusters, the signal can circulate and self-propel itself, creating a chain reaction. Determining whether the relative risk (RR) depends on the number of cells (n_nodes) will help answer the question of whether cells need to reach a sufficiently large tissue size for efficient communication.

Hypothesis: **The greater the total number of cells in the study group, the higher the relative risk (RR). Large cell populations provide more stable conditions for signal propagation.**

1.4.2 Methods

The analysis was performed on data from the `block_level_summary.csv` file.

To test whether population size affects communication, the Spearman rank correlation coefficient (ρ) between n_nodes and `relative_risk` was calculated. The analysis was performed for the control group (WT) and selected mutants.

1.4.3 Results

The analysis of the relationship between network size and signal propagation is based on two visualizations: a plot of the population size distribution (violin/boxplot) and a plot of the dependence of the relative risk on the total number of nodes (scatterplot with trend lines). Here are the main conclusions:

- Diversity of network sizes

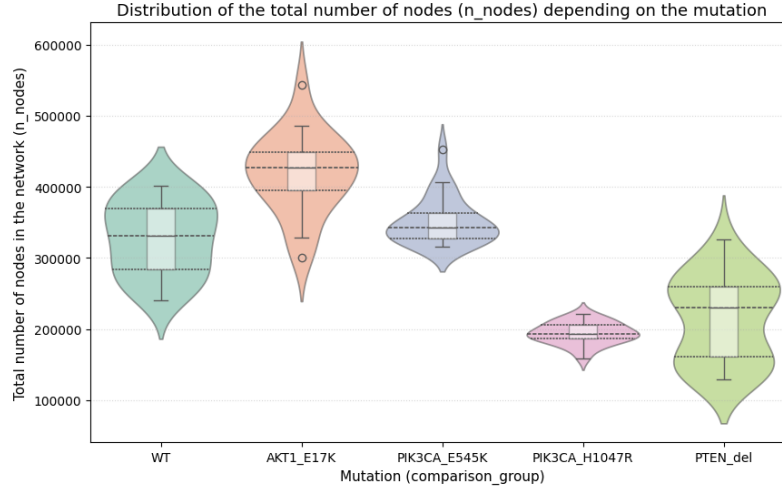


Figure 3: The violin plot reveals that individual cell lines naturally form networks of varying sizes — observe the height of the widest parts of the plot. Cells with the PIK3CA_H1047R and PTEN_del mutations form the smallest populations (typically below 250,000 nodes). In contrast, the AKT1_E17K and PIK3CA_E545K mutations dominate populations with much larger sizes (over 300,000 nodes). Wild type (WT) falls in the middle.

- The dependence of RR on network size is mutation-specific

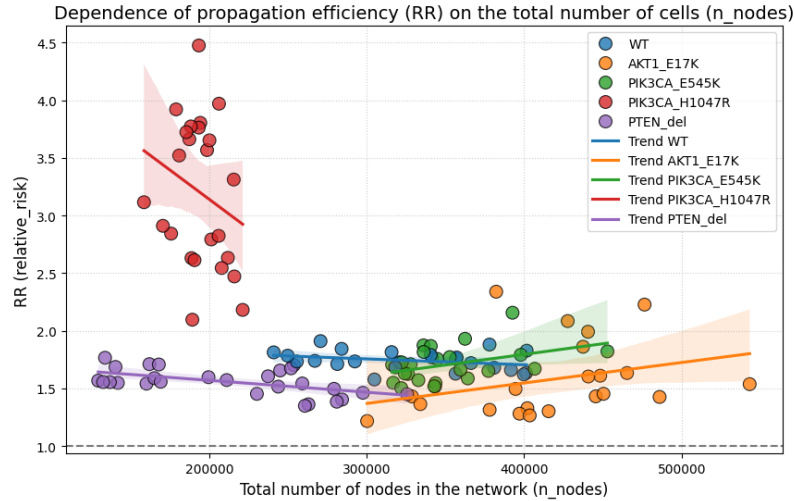


Figure 4: In the case of the control group (WT) and PTEN_del mutants, the trend lines are almost completely flat, indicating a lack of correlation between the population size and the RR value. For the AKT1_E17K and PIK3CA_E545K mutations, a positive correlation was observed — the propagation efficiency (RR) increases with the increase in the number of cells in the network. The opposite and strongest effect was observed for PIK3CA_H1047R. This mutation shows a negative correlation — it reaches extremely high RR values in the smallest populations, and with the increase in network size, its efficiency decreases.

1.4.4 Discussion

The obtained results refute the hypothesis of a universal, positive effect of population size on signal propagation, revealing substantial differences between mutations. Cells with the PIK3CA_H1047R mutation are naturally so active that they do not require a large number of neighboring cells.

The AKT1_E17K and PIK3CA_E545K mutations show the opposite behavior. Individual cells emit a relatively

weak signal, and therefore require the support of a large population for effective communication. In contrast, WT cells and PTEN_{del} mutants display stable communication efficiency.

The main limitation of this analysis is that the studied mutations naturally generate populations of very different sizes. For example, PIK3CA_{H1047R} cells occurred mainly in small groups, whereas AKT1_{E17K} cells appeared in large populations. This lack of overlap across the graph axes makes it difficult to determine whether the observed trends in RR are caused by the number of cells itself or simply reflect intrinsic properties of a given cell line. To address this limitation, future experiments should artificially control culture size, for example by forcing each mutation to grow in populations of identical size. Only such an experimental design would allow us to determine whether population size exerts a universal effect on signal propagation independently of mutation type.